

MANTHAN SHRIRAM JOSHI

Boston, MA | manthan2001@gmail.com | +1 617 735 5727 | LinkedIn | Portfolio | STEM OPT eligible, 3 years

Education

Northeastern University, Boston MA Master of Science in Cell and Gene Therapies Coursework: Cell and Gene Therapies, Advanced Drug Delivery Systems, Regulatory Landscapes, Immunology, Biotech Enterprise	December 2026
Harvard Business School, Boston MA Certificate in Management Essentials	August 2025
Manipal Institute of Technology (MAHE), Manipal India Bachelor of Technology in Biotechnology Minor: Pharmaceutical Biotechnology Four-year accredited engineering program. Coursework: Bioprocess Engineering, Downstream Processing, Bioreaction Engineering	July 2023

Skills

Process Development	Quality by Design (QbD): process characterization and criticality assessment (QTPP, CQA, CPP); deviation investigation, root-cause analysis, troubleshooting and CAPA; in-process control and acceptance criteria definition; assay system suitability (blank, full-kill control, Z'); statistical analysis and trending of assay data (dose-response fitting, %CV, signal window); process optimization and yield improvement; protocol and SOP authorship; technical report writing.
Cell Culture & Editing	Aseptic technique; mammalian cell culture (CHO) across 5+ passages at >80% viability; expansion, banking and cryopreservation; end-to-end CRISPR/Cas9 knockout workflow design; Cas9-RNP assembly with target-specific sgRNAs; CRISPRmax lipofection; cell-based cytotoxicity and kill-curve assay design (CellTiter-Glo 2.0).
Molecular & Analytical	Genomic DNA extraction; Phusion PCR; primer design in Benchling; agarose gel electrophoresis; amplicon clean-up and Nanodrop QC; Sanger sequencing and ICE indel deconvolution; restriction-enzyme digestion for construct verification; bacterial transformation, colony screening and miniprep; recombinant protein expression and Ni-NTA purification (IMAC); SDS-PAGE; Western blotting; ELISA.
CGT Domain Knowledge	CAR-T (autologous and allogeneic platforms); iPSC differentiation; lentiviral vector handling and titer determination; cGMP and CGT regulatory frameworks (coursework).
Commercial & Business	Market research and competitive landscaping; investor deck development; grant proposal support; client and stakeholder communication.
Instrumentation & Software	Bioreactor/fermenter operation; HPLC; gas chromatography; fluorescence microscopy; flow cytometry sample preparation; FlowJo; Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); ICE web tool; Microsoft Excel and Office; R (basic).

Professional Experience

Northeastern University, Teaching Assistant, Cell and Gene Therapies, Boston, MA • Taught graduate-level CGT concepts in office hours and discussion sessions, translating current CAR-T and iPSC-to-T-cell literature for students from mixed scientific backgrounds. • Reviewed student deliverables with structured written feedback and prepared literature summaries on allogeneic and autologous CAR-T for classroom case discussion.	Jan 2026 – Apr 2026
Northeastern University, Graduate Research Project, Boston, MA • Designed and executed an end-to-end CRISPR/Cas9 HPRT1 knockout workflow in CHO cells: guide selection, Cas9-RNP assembly with 2 targeting sgRNAs, CRISPRmax lipofection across 4 conditions including a non-targeting control RNP matched for nuclease and lipid load, 6-thioguanine selection, and sequence-level genotyping. • Built 8-point cell-based dose-response assays in duplicate for 6-thioguanine and puromycin on a CellTiter-Glo 2.0 readout, and specified the analytical release conditions: no-cell blank and full-kill controls, a ≥ 10 -fold signal window, and $Z' \geq 0.5$ system suitability computed before plate interpretation. • Ran the genotyping train end to end (gDNA extraction, Phusion PCR, agarose gel electrophoresis, amplicon clean-up, Nanodrop QC, bidirectional Sanger sequencing with ICE deconvolution), and set the sequencing submission gate at single-band gel identity with A260/280 of 1.7–2.0 and A260/230 ≥ 1.8 . • Ran the plasmid arm (bacterial transformation, miniprep, NotI and EcoRV diagnostic digests with gel electrophoresis, lipid transfection of an eGFP reporter at two DNA doses with fluorescence imaging for delivery efficiency and expression persistence) while maintaining CHO cultures across 5+ serial passages at >80% viability with doubling-time characterization.	Sept 2025 – Dec 2025
Reagene Biosciences Pvt Ltd, Research Intern, Bangalore, India • Built market research and competitive analysis for a 3D-bioprinted human tissue platform positioned as an alternative to animal testing in preclinical safety work. • Contributed to investor decks and grant proposals used in fundraising, translating the underlying science into a commercial case for non-technical funders. • Maintained and cryopreserved mammalian cell lines at high viability to supply the bioink formulation process for 3D-bioprinted tissue models. • Ran molecular biology and recombinant protein characterization (plasmid purification, bacterial transformation, gel electrophoresis, Nanodrop QC, Western blot, ELISA) supporting formulation development, and authored SOPs for routine lab procedures.	Aug 2023 – Aug 2024
Tranalab Pvt Ltd, Center for Human Genetics, Research Intern Bachelor's Thesis, Bangalore, India • Designed and executed a complete expression and purification process for M4, a recombinant membrane protein domain, as a bachelor's thesis project ranked first in the cohort. • Identified solubilization as the yield-limiting step and reformulated the buffer accordingly, increasing soluble protein recovery by roughly 48%. • Ran the upstream and downstream train end to end: expression in <i>E. coli</i> under IPTG induction, cell lysis by probe sonication, inclusion body solubilization, and Ni-NTA affinity chromatography (IMAC) with stepwise wash and elution. • Confirmed identity by SDS-PAGE, quantified yield by A280, buffer-exchanged by dialysis, and assessed refolding by affinity chromatography to verify retained binding function.	Jan 2023 – Aug 2023

Leadership and Projects

Process Development Report: CRISPR/Cas9 Knockout of HPRT1 in CHO Cells • Rewrote a completed CRISPR workflow as a 32-page process development document: 8 unit operations with in-process controls, quality target profile with CQA and CPP criticality assessment, 8-item deviation and CAPA analysis, an ICH Q2 fit-for-purpose qualification plan for the viability readout, and a design of experiments plan built as a 20-run resolution IV fractional factorial screen over 6 delivery parameters with a face-centered central composite follow-on. • Wrote a companion 15-page laboratory report in standard scientific format, covering assay signal window and Z' against pre-set acceptance criteria, a three-mechanism root cause assessment of the analytical method, and a ten-item corrected protocol ranked by impact.	2026
Vice President, Northeastern University, Graduate Biotech-Bioinfo Association, Boston, MA • Led executive board operations, communications and cross-committee coordination; planned networking events, workshops and industry panels.	Dec 2024 – Jan 2026
CurcuNano, Chief Executive Officer, BIOT 5219 Biotech Enterprise, Northeastern University • Led the venture team as CEO for a curcumin nanoformulation concept, owning the business case and the funding ask defended to a faculty panel; reconciled six written contributions into one consistent deck.	Spring 2026